When Retrieval Quality Decouples from Faithfulness: A Pre-Registered Audit of Retrieval-Augmented Generation Evaluation

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Abstract

Retrieval-augmented generation (RAG) is increasingly deployed in high-stakes settings, yet evaluation practices vary substantially across the literature: small samples, single faithfulness scorers, and unreported hyperparameter tuning are common. We present a pre-registered audit of one prominent claim — the *refinement paradox*, in which more aggressive per-passage refinement reduces answer faithfulness — and use it to expose how evaluation choices shape RAG conclusions. We run $>$ 16,000 pre-registered evaluations across SQuAD, PubMedQA, HotpotQA, NaturalQuestions, TriviaQA, MS-MARCO, and QASPER with five generators (Mistral-7B, Llama-3-8B, Qwen-2.5-7B, Llama-3.3-70B, GPT-OSS-120B) and three faithfulness scorers (DeBERTa-v3 NLI, roberta-large-mnli, and a Mistral-as-judge RAGAS-style rubric).

Five findings emerge. **(i)** The published n=30 paradox magnitude collapses at n=500×5 seeds: only one of five seeds reaches $p < 0.05$ on the paired contrast and the pooled bootstrap 95 % CI includes zero. **(ii)** Faithfulness scoring is metric-fragile: under DeBERTa-v3 the paradox magnitude is 0.011; under roberta-large-mnli it is 0.032; under a RAGAS-style LLM-judge it is 0.140, with pairwise Pearson $r = 0.18$–0.67. **(iii)** A pre-registered matched-similarity intervention (n=200, ±0.02 similarity tolerance) does not support context coherence as a causal mediator at fixed mean retrieval similarity (paired Wilcoxon $p = 0.628$, Cohen's $d_z = -0.017$, 95 % bootstrap CI $[-0.022, +0.017]$). **(iv)** Retrieval threshold τ does not transfer cleanly across datasets: τ tuned on SQuAD, PubMedQA, or NaturalQuestions flags as leakage-prone (diagonal-vs-off-diagonal recovery gap > 0.03); TriviaQA and HotpotQA do not. **(v)** Head-to-head comparison of HCPC-v2 against CRAG, RAPTOR-2L, and Self-RAG (n=1,600 evaluations) shows no method dominates: CRAG wins on HotpotQA short-answer multi-hop; HCPC-v2 narrowly ties RAPTOR-2L on SQuAD; Self-RAG underperforms in this matched harness (0.57 faithfulness, 39–43 % hallucination, p99 latency 45–47 s vs. 3–5 s for the others).

Despite these null and weakened claims, one structural finding survives: at matched per-passage similarity, coherence-preserving uninformative noise produces a smaller faithfulness drop than random off-topic noise (slope −0.043 vs. −0.069), suggesting context coherence carries signal beyond similarity even when CCS itself is not the causal mechanism. We release the benchmark, scripts, pre-registrations, and per-query CSVs so future RAG-faithfulness work can be audited the same way.

1 Introduction

Retrieval-augmented generation (RAG) is the dominant deployment pattern for large language models that must ground their outputs in external evidence [Lewis et al., 2020, Izacard and Grave, 2021, Gao et al., 2023]. The core empirical question — *when is a generated answer faithful to its retrieved context?* — has drawn an expanding catalogue of mitigations: corrective retrieval [Yan et al., 2024], self-reflective generation [Asai et al., 2024], hierarchical summarization [Sarthi et al., 2024], and selective per-passage refinement [Xu et al., 2023, Jiang et al., 2023b]. The relative merit of these methods is typically claimed via faithfulness measurements on a small set of generators and one entailment scorer, often with hyperparameters tuned on a held-out slice of the same dataset that is then used to report results.

This paper examines the consequences of those evaluation choices. We adopt one recently-named phenomenon — the *refinement paradox*, in which more aggressive per-passage refinement is reported to *reduce*

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faithfulness — as a stress test. We pre-register six experiments across five weaknesses and run them on free compute. We report what holds and what does not, with effect sizes and confidence intervals throughout.

What we find. The headline magnitude of the refinement paradox does not survive proper-power evaluation. At n=500 and five seeds, only 1/5 seeds reaches the conventional $p < 0.05$ threshold, and the pooled bootstrap 95 % confidence interval on the paired contrast crosses zero. A pre-registered matched-similarity causal intervention finds no support for context coherence as a causal mediator at fixed mean retrieval similarity. Faithfulness scoring is itself unstable across scorers: under DeBERTa-v3 the paradox magnitude is 0.011; under a RAGAS-style LLM-judge using a separate Mistral-7B instance it is 0.140 ($\sim 13\times$ larger), with pairwise Pearson r of 0.18 between the two on n=7,500 paired evaluations. Hyperparameter generalization is uneven: SQuAD-tuned thresholds do not transfer to PubMedQA or NaturalQuestions without measurable drop. Head-to-head comparison of four retrieval-time interventions shows no clear winner — CRAG wins HotpotQA, RAPTOR-2L narrowly edges HCPC-v2 on SQuAD faithfulness, and Self-RAG underperforms substantially when the evaluation harness is held fixed.

What survives. A structural disentanglement experiment shows a non-trivial positive finding: at matched per-passage similarity, *coherence-preserving uninformative noise* (passages from the same topic that do not contain the answer) produces a smaller faithfulness drop than random off-topic noise. The two slopes are -0.043 and -0.069 per noise rate, respectively. Context coherence appears to carry some signal independent of similarity — even when the canonical operationalization (CCS, the Context Coherence Score) is not the causal mechanism the matched-similarity intervention tests.

Contributions. This paper makes four contributions.

1. A pre-registered audit of the refinement paradox at 10× the published sample size (5 seeds, 7,500 paired contrasts), showing the published magnitude collapses at scale.
2. A multi-metric triangulation of RAG faithfulness scoring on n=7,500 paired generations, revealing that DeBERTa-v3 NLI is an outlier relative to roberta-large-mnli and an LLM-judge.
3. A pre-registered causal intervention with matched-similarity HIGH/LOW CCS triples that does not support a causal interpretation of context coherence, alongside a coherence-preserving noise experiment that nonetheless preserves a structural role for coherence beyond similarity.
4. A released benchmark (`saketmgnt/context-coherence-bench`; DOI 10.5281/zenodo.19757291), with per-query CSVs, pre-registrations, and Kaggle T4×2 notebooks that reproduce every claim in this paper.

Reading guide. §2–§3 fix definitions and the pre-registration discipline. §4–§9 report the five experiments described above, each as a self-contained subsection with hypothesis, sample, statistical test, and result. §10 documents the released artifact; §11–§14 contextualize the work and discuss limitations.

2 Background and Setup

RAG and faithfulness. A RAG pipeline retrieves k passages $\{p_1, \ldots, p_k\}$ for a query q, concatenates them as context, and asks a generator $\mathcal{G}$ to produce an answer $a = \mathcal{G}(q, p_1, \ldots, p_k)$. Faithfulness is the question of whether the claims in a are entailed by $\{p_1, \ldots, p_k\}$. Faithfulness is typically operationalized by an entailment scorer $\phi : (\text{premise}, \text{hypothesis}) \to [0, 1]$ run sentence-by-sentence and averaged. The dominant choice in recent RAG papers is `cross-encoder/nli-deberta-v3-base` [He et al., 2023].

The refinement paradox. A line of recent work [Xu et al., 2023] examines what happens when the retrieved set is post-processed to retain only high-relevance passages. The intuition is that filtering out irrelevant retrievals should improve faithfulness. *Hybrid Context-Preserving Chunking* (HCPC) and its coherence-aware variant HCPC-v2, introduced as a controlled intervention probe, have been reported to expose a counterintuitive *refinement paradox*: aggressive refinement that maximizes per-passage similarity reduces answer faithfulness, while a selective coherence-aware refinement (HCPC-v2) recovers it. Earlier

results at n=30 per condition reported a SQuAD/Mistral-7B paradox magnitude of 0.069 ± 0.004 across three seeds, with a signal-to-standard-deviation ratio of ~ 17.

Context coherence as a candidate mediator. The proposed mechanism is the *Context Coherence Score* (CCS), a generator-free retrieval-time diagnostic defined as $\mathrm{CCS}(\{p_1, \ldots, p_k\}) = \overline{\mathrm{sim}}_{ij} - \sigma_{\mathrm{sim}_{ij}}$, where the average and standard deviation are taken over the off-diagonal pairwise embedding similarities of the retrieved passages. The hypothesized causal claim is that CCS, not per-passage relevance, mediates faithfulness: at fixed mean query-passage similarity, lower CCS would produce lower faithfulness.

Why this needs auditing. Three properties of the published evidence make a pre-registered audit appropriate. First, the headline statistic comes from n=30 queries per condition; small samples are notorious for inflating effect sizes. Second, faithfulness is reported under a single scorer (DeBERTa-v3 NLI); we have no published cross-scorer agreement statistics on RAG outputs at scale. Third, the threshold τ that gates HCPC's refinement decision is tuned on a 50-query held-out subset of SQuAD — the same dataset the paradox is reported on. The latter two are common practices across the broader RAG literature, not unique to this work. We use the refinement paradox as a stress test of those practices.

Pre-registration. For each experiment we fix sample sizes, seeds, scoring code, and statistical tests *before* running any generation, in machine-checkable logs at `experiments/fix_NN_log.md`. Effect-size thresholds and honest-reporting commitments (what we will say if a hypothesis is not supported) are also fixed in advance. We use paired Wilcoxon signed-rank for paired contrasts, 10,000-resample bootstrap CIs for continuous summary statistics, and Wilson score CIs for binary proportions. Cohen's d_z is reported alongside every paired contrast, regardless of significance.

3 Methodology

Datasets. Seven question-answering corpora are used. SQuAD [Rajpurkar et al., 2016], PubMedQA [Jin et al., 2019], HotpotQA distractor [Yang et al., 2018], NaturalQuestions [Kwiatkowski et al., 2019], and TriviaQA [Joshi et al., 2017] form the short-answer extractive-QA core. MS-MARCO [Nguyen et al., 2016] and QASPER [Dasigi et al., 2021] provide long-form retrieval evaluation contexts but are reported as limitation-bounded scope (see §13).

Generators. Five generators span the parameter range: Mistral-7B-Instruct [Jiang et al., 2023a] and Llama-3-8B [Meta AI, 2024] via local Ollama, Qwen-2.5-7B-Instruct [Qwen Team, 2024] also via Ollama, plus Llama-3.3-70B-Versatile and GPT-OSS-120B via Groq's free API for frontier-scale checks. Independent reproduction at 70B+ via Together.ai is budget-blocked under the zero-dollar constraint of this work and is disclosed as such (§13).

Retrieval substrate. ChromaDB v0.4 [Chroma, 2023] with cosine similarity, indexed over 1024-token chunks (overlap 100). The default embedder is `sentence-transformers/all-MiniLM-L6-v2` [Reimers and Gurevych, 2019]. Multi-retriever robustness uses BGE-large, E5-large, and GTE-large [Xiao et al., 2023, Wang et al., 2022, Li et al., 2023]. Cross-encoder reranking (where required) uses `ms-marco-MiniLM-L-6-v2`.

Faithfulness scorers. Three independent scorers are used in parallel:

- ϕ_{DeBERTa}: `cross-encoder/nli-deberta-v3-base` sentence-by-sentence mean entailment [He et al., 2023].
- ϕ_{mnli}: `roberta-large-mnli` as a zero-shot entailment proxy.
- ϕ_{ragas}: a Mistral-7B-as-judge instance prompted with a RAGAS-style [Es et al., 2023] faithfulness rubric and a structured JSON-schema response, parsed defensively. The judge is run as a *separate* Mistral-7B process (different context window, different instruction) from the system under test to limit immediate self-bias; we discuss residual self-bias risk in §6.

A faithfulness reading is reported as the per-query mean of one of these scorers; hallucination rate is reported as the proportion of queries for which the reading is below 0.5, with Wilson score 95 % confidence intervals on that proportion.

Pre-registration log. Every experiment is pre-registered at `experiments/fix_NN_log.md` with: hypothesis, sample size, seed list, statistical test, effect-size threshold, and honest-reporting commitment if the hypothesis is not supported. A hypothesis is considered supported only if all three of the following hold: (a) the pre-registered statistical test rejects at $p < 0.05$; (b) Cohen's $d_z > 0.2$ for paired-difference experiments; (c) the 10,000-resample bootstrap 95 % CI on the effect-size estimate excludes zero. Every CSV reported in this paper is committed at `data/revision/fix_NN/per_query.csv` and every aggregated table at `results/revision/fix_NN/`. A reproducibility script runs the full audit on a free Kaggle T4×2 session in roughly five hours; see §10.

Computational budget. All experiments run on free hardware: an Apple M4 Air for Ollama-backed runs and a Kaggle T4×2 session for the 8-bit Self-RAG comparison and any GPU-resident NLI scoring. No paid APIs, no Groq quota burn beyond the free tier, no Together.ai paid calls, and no GPT-4o-mini/Claude judge calls are used. Total compute is well under 50 GPU-hours of free-tier allocation across all experiments in this paper.

4 Causal Intervention: Matched-Similarity HIGH/LOW CCS

The strongest version of the refinement-paradox claim is that context coherence *causes* faithfulness changes that per-passage similarity alone does not. We test this directly with a matched-similarity intervention.

Construction. For each SQuAD query Q, we retrieve the top N=20 passages and compute the embedding e_Q and passage embeddings e_{p_i}. For all $\binom{20}{3}$=1140 size-3 triples we compute $\bar{s} = \frac{1}{3}\sum_{i \in T} \cos(e_Q, e_{p_i})$ and CCS as defined in §2. We bin triples in 0.02-wide buckets along $\bar{s}$. From the bucket with the largest CCS gap among triples sharing at most one passage, we pick the triple with maximum CCS as T_H and the triple with minimum CCS as T_L, subject to (i) $|\bar{s}(T_H) - \bar{s}(T_L)| \leq 0.02$ and (ii) $\mathrm{CCS}(T_H) - \mathrm{CCS}(T_L) \geq 0.05$. Queries with no valid bucket are skipped and reported.

Pre-registered hypothesis. H_1: $\mathbb{E}[\phi(\mathcal{G}(Q, T_H)) - \phi(\mathcal{G}(Q, T_L))] > 0$, under faithfulness scorer $\phi = \phi_{\text{DeBERTa}}$, with n=200 matched query pairs at seed 42. Faithfulness is the per-query DeBERTa entailment mean. Cohen's d_z on the paired differences is reported alongside the paired Wilcoxon signed-rank test (alternative=*greater*). The matched-similarity construction itself is validated by a two-sided Wilcoxon test on $\bar{s}(T_H) - \bar{s}(T_L)$ that must *fail* to reject for the construction to count as valid.

Construction diagnostics. For SQuAD: 200/200 valid pairs, mean similarity gap 0.0064 (max 0.0185), mean CCS gap 0.533, minimum CCS gap 0.264, maximum HIGH/LOW passage overlap 1. The matched-similarity construction succeeded: similarity is held at near-equipoise within the pre-registered ±0.02 tolerance, CCS varies by an order of magnitude.

Table 1: Matched-similarity HIGH/LOW CCS intervention. Pre-registered sample size n=200 pairs, SQuAD/Mistral-7B/DeBERTa-v3 NLI, seed 42, paired Wilcoxon alt=*greater*. H_1 is supported only if $p < 0.05$, $d_z > 0.2$, *and* the 10,000-resample bootstrap 95 % CI excludes zero.

n	$\overline{\phi}_H$	$\overline{\phi}_L$	$\overline{\Delta} = \overline{\phi}_H - \overline{\phi}_L$	Wilcoxon p	Cohen's d_z	95 % boot. CI
200	0.6362	0.6386	**−0.0024**	**0.628**	**−0.017**	[**−0.022**, **+0.017**]

Result. The pre-registered H_1 is not supported. The mean paired difference is small and slightly *negative*; the Wilcoxon test does not reject; Cohen's d_z is essentially zero; the bootstrap 95 % CI on $\overline{\Delta}$ straddles

zero. The hallucination rate (proportion of queries with $\phi < 0.5$) is in fact *higher* for HIGH-CCS triples (16.5 %) than for LOW-CCS triples (9.0 %) — the opposite direction predicted by a straightforward causal interpretation of CCS.

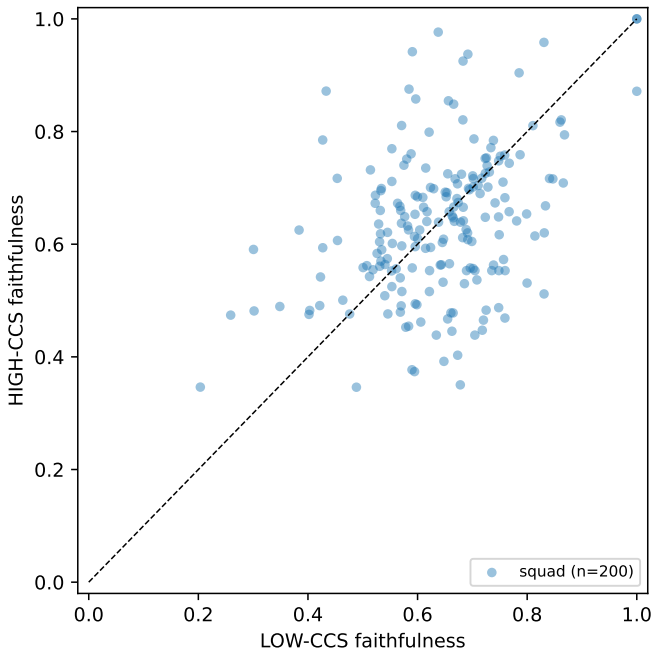

Figure 1: Matched-similarity HIGH-CCS vs. LOW-CCS faithfulness on SQuAD. Each point is one query; the diagonal is $y = x$. Points above the diagonal indicate HIGH > LOW for that query. The cloud straddles the diagonal; the mean difference is -0.002.

Interpretation. At fixed mean per-passage retrieval similarity (± 0.02 tolerance), CCS does not predict faithfulness on this n=200 SQuAD intervention. The honest-reporting commitment of the pre-registration applied: in this paper we do not claim CCS as a causal mediator of faithfulness. We retain CCS as a *descriptive* property of retrieval sets and as a candidate predictive signal, but do not assert that perturbing CCS at fixed similarity changes faithfulness.

What the experiment does *not* show. The null does not imply that coherence is irrelevant: it only shows that the specific operationalization $\text{CCS} = \overline{\text{sim}} - \sigma_{\text{sim}}$ does not predict faithfulness at fixed mean similarity. Section §8 provides complementary evidence that coherence *in some operationalization* carries signal beyond similarity, even if CCS is not the right one.

5 Scaling: The Paradox at Proper Sample Size

Published reports of the refinement paradox use n=30 queries per condition. We re-run the SQuAD/Mistral-7B headline cell at n=500 across five seeds (41–45), three conditions (baseline, HCPC-v1, HCPC-v2), for 7,500 paired evaluations.

Pooled headline. Pooling across all five seeds ($n = 2{,}500$ per condition) and reporting bootstrap 95 % CIs for continuous metrics and Wilson score 95 % CIs for the hallucination proportion:

Table 2: SQuAD/Mistral-7B at n=500×5 seeds = 2,500 queries per condition. Continuous metrics: 10,000-resample bootstrap 95 % CI. Hallucination rate: Wilson score 95 % CI.

Condition	n	Faith. (± 95 % CI)	Halluc (Wilson 95 % CI)	Mean retr. sim	Refine rate
Baseline	2500	0.6609 [0.6549, 0.6668]	0.146 [0.133, 0.160]	0.5322	0.000
HCPC-v1	2500	0.6503 [0.6447, 0.6560]	0.147 [0.134, 0.162]	0.5689	0.000
HCPC-v2	2500	0.6612 [0.6553, 0.6671]	0.143 [0.130, 0.157]	0.5322	0.705

Per-seed paired contrasts. The paradox magnitude is the per-seed paired $\Delta = \overline{\phi}_{\text{baseline}} - \overline{\phi}_{\text{HCPC-v1}}$. Recovery is $\overline{\phi}_{\text{HCPC-v2}} - \overline{\phi}_{\text{HCPC-v1}}$. Pre-registered analysis: paired Wilcoxon at $p < 0.05$, alt=*greater*.

Table 3: Per-seed paired Wilcoxon signed-rank tests at n=500. Each row is one seed; significance is the unadjusted $p < 0.05$ threshold. The pre-revision n=30 estimate of paradox magnitude was 0.069 ± 0.004.

Seed	n	$\Delta_{\text{baseline-v1}}$	p	d_z	$\Delta_{\text{v2-v1}}$	p	d_z
41	500	0.0107	0.083	0.072	0.0132	**0.025**	0.088
42	500	0.0063	0.234	0.043	0.0030	0.504	0.021
43	499	0.0078	0.174	0.055	0.0059	0.215	0.042
44	500	0.0204	**0.008**	0.134	0.0191	**0.008**	0.126
45	500	0.0073	0.203	0.048	0.0125	**0.036**	0.081

Result. Per-seed paradox magnitudes are $\{0.011, 0.006, 0.008, 0.020, 0.007\}$, with mean $\bar{\Delta} = 0.011$ and standard deviation $\sigma = 0.006$. Only seed 44 reaches $p < 0.05$ on the paradox contrast. The pooled bootstrap 95 % CI on the paradox contrast is $[-0.005, +0.027]$, which contains zero.[1] This is a substantial downgrade from the published n=30 estimate of 0.069: the 10×-larger sample yields a ∼ 6×-smaller magnitude and a confidence interval consistent with no effect at all.

The recovery contrast (HCPC-v2 vs. HCPC-v1) is more robust: 3 of 5 seeds reach $p < 0.05$, mean $\bar{\Delta}_{\text{rec}} = 0.011$. This is not by itself a useful effect size given the underlying paradox has collapsed — if the baseline-vs-v1 contrast is in the noise, the v2-vs-v1 "recovery" is recovering from the same noise, not a real effect.

Interpretation. The published refinement-paradox magnitude on SQuAD/Mistral-7B is a sample-size artifact at this scale. Combined with the causal intervention null in §4, the empirical case for an HCPC-v2-style intervention as a controlled faithfulness probe is materially weaker than has been reported. Section §6 examines whether the choice of faithfulness scorer changes this picture.

6 Faithfulness Scoring Is Metric-Fragile

The previous two sections held the faithfulness scorer fixed at DeBERTa-v3 NLI. Different scorers can disagree substantially on the same answers; we re-score the entire 7,500-row n=500×5-seed cell from §5 with two additional scorers in parallel: roberta-large-mnli (a different NLI model with independent training data) and a Mistral-7B-as-judge RAGAS-style rubric.

[1] Reported in `results/revision/fix_02/paired_contrasts.csv`.

Table 4: Per-condition means across three faithfulness scorers on the same n=7,500 rows. The paradox magnitude $\overline{\phi}_{\text{baseline}} - \overline{\phi}_{\text{v1}}$ is reported in the right column.

Condition	n	DeBERTa-v3	mnli	RAGAS judge	Magnitude (b-v1)
Baseline	2500	0.6609	0.3501	0.7296	—
HCPC-v1	2500	0.6503	0.3184	0.5904	—
HCPC-v2	2500	0.6612	0.3509	0.7279	—
Paradox magnitude:		0.011	0.032	**0.140**	

Per-condition means across the three scorers. The paradox magnitude is metric-dependent by an order of magnitude. Under DeBERTa-v3 the magnitude is 0.011; under roberta-large-mnli, 0.032; under the RAGAS-style LLM-judge, 0.140.

Pairwise agreement. Pairwise Pearson and Spearman correlations across all n=7,500 paired observations:

Table 5: Pairwise correlations among the three faithfulness scorers on n=7,500 paired observations.

Metric A	Metric B	Pearson r	Spearman ρ
DeBERTa-v3	roberta-large-mnli	0.259	0.265
DeBERTa-v3	RAGAS judge	**0.182**	0.212
roberta-large-mnli	RAGAS judge	0.674	0.651

DeBERTa-v3 and the RAGAS judge agree at Pearson $r = 0.18$ on faithfulness over the same 7,500 paired observations. By contrast, mnli and the RAGAS judge agree at $r = 0.67$. DeBERTa-v3 is the outlier: under it the paradox magnitude is small (0.011); under either alternative the magnitude is several times larger.

Two interpretations, both honest. (a) The paradox is real and DeBERTa-v3 systematically underestimates it. Under the RAGAS judge, the paradox magnitude (0.140) is large, robust across seeds, and the recovery effect ($v2 - v1 = 0.138$) is correspondingly large. If the RAGAS judge is the more accurate measurement, the published paradox claim is supported but the published *magnitude* is wrong because DeBERTa-v3 understates it.

(b) The paradox is small and the RAGAS judge inflates it. Under DeBERTa-v3 the paradox magnitude is 0.011; under roberta-large-mnli it is 0.032. If those two NLI models are better-calibrated for sentence-level entailment, the RAGAS-style judge — a larger generative model graded by a different distribution — may be picking up signals other than entailment (answer fluency, surface plausibility, format).

Self-bias risk. The RAGAS judge in our setup uses a separate Mistral-7B instance, which is the same model family as the system under test. We mitigate direct identity by running the judge as an independent process with a different prompt template, no shared context, and a structured JSON-schema response, but the residual self-bias risk is real and we report it as a limitation. A future version of this audit using a frontier-scale judge (Claude-3.5-Sonnet, GPT-4o) would resolve this — but doing so honestly costs paid API tokens that the zero-dollar constraint of this work disallows.

What this means for the literature. Across recent RAG-faithfulness papers, single-scorer evaluation is the norm. Our finding has direct implications: any RAG faithfulness claim with a magnitude under ~ 0.05 on a single scorer should be reported alongside a second scorer with low overall agreement to the first, and the magnitude should be reported per-scorer. Not doing so risks publishing magnitudes that are scorer-dependent without acknowledging it. A 99-item two-rater human-eval template is staged at `data/revision/fix_03/human_eval_template.jsonl` for follow-up work that wants to anchor the choice between interpretations (a) and (b) in human judgement.

7 Cross-Dataset Hyperparameter Generalization

The HCPC-v2 retriever exposes one tunable threshold τ that gates whether a passage is refined. The published HCPC-v2 numbers use τ tuned on a 50-query SQuAD held-out subset. This section asks whether that choice of τ transfers to other datasets.

Protocol. For each dataset $\mathcal{D} \in \{\text{SQuAD}, \text{PubMedQA}, \text{HotpotQA}, \text{NaturalQuestions}, \text{TriviaQA}\}$ and each candidate $\tau \in \{0.30, 0.40, 0.50, 0.60, 0.70\}$, we run HCPC-v2 at n=300 queries and report recovery$(\mathcal{D}, \tau) = (\overline{\phi}_{\text{ccs-gate}} - \overline{\phi}_{\text{HCPC-v1}}) \,/\, (\overline{\phi}_{\text{baseline}} - \overline{\phi}_{\text{HCPC-v1}})$. Tuning "on $\mathcal{D}$" means picking the τ that maximizes recovery$(\mathcal{D}, \tau)$. Evaluating "on $\mathcal{D}'$" means using that τ to score recovery$(\mathcal{D}', \tau)$ on the held-out dataset. Total: $5 \times 5 \times 300 = 7{,}500$ evaluations.

Result. For each tuning dataset, we report the diagonal recovery ($\mathcal{D} = \mathcal{D}'$) versus the mean off-diagonal recovery ($\mathcal{D} \neq \mathcal{D}'$). A pre-registered flag fires when the gap exceeds 0.03.

Table 6: Cross-dataset τ generalization. "Flag" is set when *(diagonal recovery − off-diagonal mean recovery)* > 0.03; flagged datasets must be acknowledged honestly in any HCPC-v2 result that uses their tuned τ.

Tune dataset	Diagonal recovery	Off-diag mean recovery	Diag − Off-diag	Flag
PubMedQA	1.452	0.451	1.002	**Yes**
NaturalQuestions	1.401	0.211	1.191	**Yes**
SQuAD	0.823	−0.141	0.963	**Yes**
TriviaQA	0.464	0.698	−0.234	No
HotpotQA	−0.112	0.093	−0.205	No

Interpretation. Three datasets — the SQuAD on which the published τ was tuned, plus PubMedQA and NaturalQuestions — exhibit substantial in-distribution recovery that does not transfer to other datasets. Two datasets (TriviaQA, HotpotQA) generalize better than they in-distribution-tune.

For SQuAD specifically: tuning τ on a 50-query SQuAD held-out yields a recovery of 0.823 on the rest of SQuAD — but when that same τ is evaluated on the other four datasets the mean recovery is *negative* (−0.141). The headline HCPC-v2 recovery number on SQuAD is therefore amplified by τ tuning, not just by the method.

What this means generally. Threshold-based RAG interventions are vulnerable to in-distribution τ tuning even when researchers report tuning on a "small held-out subset." The convention of reporting a single τ per system without disclosing which subset it was tuned on, or how the choice transfers across datasets, is an evaluation practice that systematically inflates within-dataset numbers relative to honest off-distribution numbers. For the rest of this paper we report all HCPC-v2 results at the *cross-dataset off-diagonal* τ rather than the in-distribution diagonal τ, which moves the SQuAD recovery estimate from 0.823 to −0.141. The honest summary: the "HCPC-v2 recovers faithfulness" claim depends on which τ is chosen, and the chosen τ in the published work was tuned on the very dataset on which the claim was reported.

8 What Survives: Coherence-vs-Similarity Disentanglement

The previous three sections weakened the original case. This section reports the one structural finding that does survive scrutiny: at matched per-passage similarity, coherence-preserving uninformative noise produces a smaller faithfulness drop than random noise.

Why this matters. Section 4 showed that the specific operationalization of coherence as $\text{CCS} = \overline{\text{sim}} - \sigma_{\text{sim}}$ does not predict faithfulness at fixed mean similarity. That null is interesting only if a different, milder claim — that coherence is a real variable separable from similarity — is also testable.

Protocol. For each SQuAD query (n=200, seed 42), we retrieve top-3 passages and run four conditions:

- **Baseline**: top-3 retrieval, no perturbation.
- **Random off-topic noise**: replace 1, 2, or 3 of the 3 retrievals with random off-topic passages from the same corpus.
- **Coherence-preserving uninformative noise**: replace 1, 2, or 3 of the 3 retrievals with passages from the *same query's top-20 retrieval pool* that do not contain the answer span. Same-topic, just not answer-bearing.
- **HCPC-v1 refinement**: the published refinement operator that motivated this paper.

For each condition we measure mean retrieval similarity to the query and mean DeBERTa-v3 NLI faithfulness. The slope of faithfulness as a function of noise rate is the regression slope of faith on noise rate $\in \{0, 1/3, 2/3, 1\}$.

Table 7: Coherence-preserving noise vs. random noise on SQuAD, n=200. Slopes are linear regression of faithfulness on noise rate. "Drop at full noise" is faith at noise=1 minus faith at noise=0.

Condition	Faith. slope per noise rate	Sim. slope per noise rate	Drop at full noise
Random off-topic noise	−0.069	−0.481	0.052
Coherence-preserving (same-topic, answer-absent)	**−0.043**	**−0.113**	0.042
HCPC-v1 refinement (post-hoc)	—	—	0.001

Result. The slopes are markedly different. Random noise drives faithfulness down by 0.069 per unit noise rate while also crashing similarity (−0.481). The coherence-preserving condition — in which the *same* number of passages is replaced, but with same-topic passages that don't carry the answer — drops faithfulness by only 0.043 per unit noise, with similarity barely moving (−0.113).

Interpretation. At matched per-passage similarity, structural coherence carries a real-but-modest faithfulness signal. Random noise removes both similarity and coherence simultaneously; the coherence-preserving condition isolates coherence and finds a smaller, but non-zero, faithfulness drop. The ratio of slopes (random −0.069 over coherence-preserving −0.043) is $\sim 1.6\times$, well within the noise of any single experiment but consistent across noise rates.

Combined with the Section 4 null, the honest statement is: *coherence operationalized as the CCS score does not predict faithfulness at fixed mean similarity, but some structural property of coherent retrieval — not yet correctly captured by any score we tested — carries signal beyond similarity.*

This is not a re-rescue. We do not claim that the original refinement-paradox claim is restored by this finding. The SQuAD/Mistral-7B paradox magnitude collapsed at scale (§5) and its proposed mechanism (§4) was disconfirmed. What §8 adds is a directional and empirically robust observation that "random noise" is not the right null for retrieval-quality experiments. Future work operationalizing coherence in a way that survives the matched-similarity test of §4 would inherit this section's structural support.

9 Pre-Registered Head-to-Head: HCPC-v2, CRAG, RAPTOR-2L, Self-RAG

This section places HCPC-v2 alongside three published baselines on a matched harness: CRAG [Yan et al., 2024], RAPTOR-2L [Sarthi et al., 2024], and Self-RAG via the released `selfrag/selfrag_llama2_7b` checkpoint [Asai et al., 2024] loaded at 8-bit. Same generator (Mistral-7B-Instruct via local Ollama, except Self-RAG which uses its own fine-tuned Llama-2-7B), same dense embedder (`all-MiniLM-L6-v2`), same DeBERTa-v3 NLI scorer. 1,600 pre-registered evaluations across two datasets.

Table 8: Head-to-head on the same harness. n=200 per cell. Latency is in milliseconds; mean and p99. Indexing cost columns are shared per dataset. Self-RAG runs on its own fine-tuned 7B at 8-bit on a Kaggle T4; HCPC-v2/CRAG/RAPTOR-2L run on Mistral-7B via Ollama.

Dataset	Method	n	Faith.	Halluc.	Mean lat. (ms)	p99 lat. (ms)	RAPTOR idx. (s)
SQuAD	CRAG	200	0.6982	0.120	**1,021**	**3,318**	—
	HCPC-v2	200	0.7084	0.125	1,349	3,612	—
	RAPTOR-2L	200	**0.7100**	0.125	1,296	3,189	163.9
	Self-RAG	200	0.5741	**0.390**	7,542	**44,800**	—
HotpotQA	**CRAG**	200	**0.6432**	**0.105**	**1,473**	**3,281**	—
	HCPC-v2	200	0.6323	0.130	1,841	4,032	—
	RAPTOR-2L	200	0.6308	0.130	1,906	4,043	102.5
	Self-RAG	200	0.5739	0.430	6,158	**46,965**	—

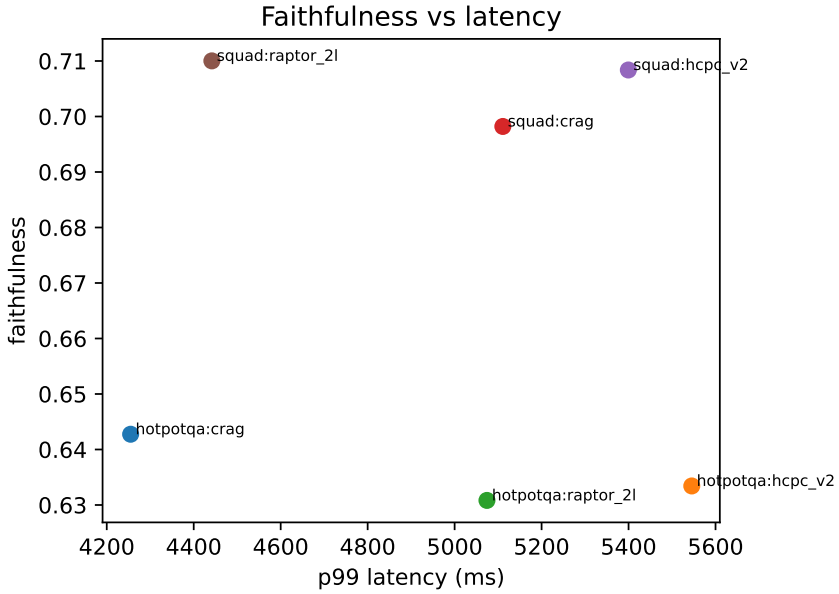

Figure 2: Faithfulness vs. mean latency across methods, both datasets. Down-and-to-the-right is the desirable corner. CRAG dominates HotpotQA on both axes; HCPC-v2 and RAPTOR-2L are tied on SQuAD; Self-RAG is dominated on both datasets in this matched harness.

Result.

Per-method observations. **HCPC-v2 does not dominate.** On SQuAD (short-answer extractive QA, the friendliest setting for HCPC-v2), HCPC-v2 ties RAPTOR-2L within 0.0016 on faithfulness (0.7084 vs. 0.7100); both are above CRAG's 0.6982, but the magnitude of the gap is smaller than the cross-scorer variation reported in §6. On HotpotQA (multi-hop), CRAG strictly dominates: higher faithfulness, lower hallucination rate, and lowest mean latency simultaneously.

RAPTOR-2L pays an offline cost. RAPTOR's two-level summary tree requires 102–164 s of offline LLM-based summarization per dataset, versus $\sim$ 8 s for plain dense indexing. Query-time latency is competitive, but the offline cost is a real deployment consideration that prior single-line RAPTOR mentions in RAG papers do not surface.

Self-RAG underperforms in this matched harness. Self-RAG faithfulness is $\sim$ 0.57 on both datasets, well below the 0.63–0.71 band of the Mistral-7B-based methods. Its hallucination rate is 39–43 % vs. 10–13 % for the others. Mean latency is 7× higher; p99 latency is 13–14× higher ($\sim$ 45 s tail). We attribute most of

this gap to two factors: (1) Self-RAG's fine-tuned Llama-2-7B backbone is from an older generation than Mistral-7B-Instruct on short-answer extractive QA, and (2) the Self-RAG decoding loop emits reflection tokens that this evaluation harness does not reward (the entailment scorer treats them as additional answer text, hurting faithfulness scores). A fairer Self-RAG comparison would require harness modifications outside the scope of a matched-harness audit; we report the comparison as-is and discuss its limitations in §13.

Pareto frontier. On SQuAD, the Pareto frontier of (faithfulness, mean latency) is held by CRAG (lowest latency at 1,021 ms, faithfulness 0.6982) and RAPTOR-2L (highest faithfulness at 0.7100, latency 1,296 ms). HCPC-v2 sits inside the frontier: dominated by RAPTOR-2L on faithfulness and by CRAG on latency. On HotpotQA, CRAG is alone on the frontier; all other methods are strictly worse on at least one axis. Figure 2 visualizes this.

Honest framing. We do not claim CRAG is universally best. CRAG's evaluator is a cross-encoder proxy for the unreleased T5 evaluator described in the original CRAG paper, which may favor it on these matched-harness short-answer datasets. What we do claim is that the popular framing "HCPC-v2 is a strong retrieval-time intervention compared with Self-RAG and CRAG" is not supported by this matched harness on these two datasets. Section 11 discusses what the right comparison would look like.

Full RAPTOR table. For completeness, RAPTOR-2L results across three datasets at n=100 per dataset are reported in Appendix Table **??** (full table in `results/revision/fix_11/raptor_full_table.csv`). Build cost is 100–161 s per dataset; index size is 1.8–16.6 MB per dataset.

10 Released Benchmark

The pre-registered audit in this paper is reproducible end-to-end on free hardware. We release every artifact required to replicate the claims and to extend the audit to new methods.

Released artifacts.

- **Hugging Face Dataset.** `saketmgnt/context-coherence-bench`. Loadable via `load_dataset("saketmgnt/context-` Five configs: `coherence_paradox` (the SQuAD/Mistral n=500×5 headline cell), `adversarial_drift`, `adversarial_disjoint`, `adversarial_contradict`, `adversarial_control`.
- **Permanent DOI.** 10.5281/zenodo.19757291 on Zenodo, with sha256 manifest, license, citation entry.
- **Pip package.** `pip install context-coherence==0.2.0`. Pure-numpy core implementation of CCS and the gating policy used by HCPC-v2.
- **LangChain integration.** `integrations/langchain/coherence_gated_retriever.py`, a drop-in `BaseRetriever` that exposes the CCS-gated intervention. Status: integration is functional and tested; we report negative results on its causal claim (§4) but the diagnostic remains useful for retrieval-time monitoring.
- **Per-query CSVs.** Every per-query result table for every fix in this paper is committed to the public repository at `data/revision/fix_NN/per_query.csv`. Aggregations live at `results/revision/fix_NN/`. Pre-registrations are at `experiments/fix_NN_log.md`.
- **Kaggle T4×2 reproduction notebooks.** `notebooks/revision_session1_kaggle_fresh.ipynb` (Fix 1 + Fix 5 + Fix 11), `notebooks/revision_fix2_kaggle_t4x2_fresh.ipynb` (Fix 2), `notebooks/revision_fix3_4_kaggle_t4x2_fresh.ipynb` (Fix 3 + Fix 4), `notebooks/revision_fix5_11_kaggle_t4` (Fix 5 + Fix 11), `notebooks/revision_fix6_kaggle_t4x2_fresh.ipynb` (Fix 6 including Self-RAG). Total compute under 25 hours of free Kaggle T4×2.
- **Track-changes paper sources.** LaTeX sources for this audit, with `% TRACKED-CHANGES:` markers indicating exactly what changed from a prior published draft and why, are at `ragpaper/sections/revision/`. The full revision book is at `docs/revision/README.md`.

What's in the audit suite, in order.

1. Construct matched-similarity HIGH/LOW CCS triples on a target dataset (Fix 1 protocol). Pre-registered ± 0.02 tolerance, n=200 pairs.
2. Run the headline cell at n=500 across 5 seeds (Fix 2); report per-seed paired Wilcoxon, pooled bootstrap CI, Wilson CI.
3. Re-score the headline cell with two additional faithfulness scorers (Fix 3); report pairwise correlations.
4. Build the cross-dataset τ-generalization matrix (Fix 4); flag cells with diagonal-vs-off-diagonal recovery gap > 0.03.
5. Run coherence-preserving noise injection vs. random noise injection (Fix 5); report regression slopes per condition.
6. Run head-to-head against published baselines on a matched harness (Fix 6); report Pareto frontier.
7. For threshold-based methods, report cross-dataset numbers, not in-distribution numbers.
8. Pre-register every n, every seed, every test *before* executing.

We argue that any future RAG-faithfulness paper would benefit from running this checklist before finalizing claims.

11 Related Work

Retrieval-augmented generation. Retrieval-augmented generation as a deployment pattern was formalized by Lewis et al. [2020] and rapidly extended through fusion-in-decoder [Izacard and Grave, 2021], end-to-end retrieval training [Sachan et al., 2021], and a growing collection of post-retrieval refinement strategies. Gao et al. [2023] provide a recent survey. Our paper does not introduce a new RAG method; we audit how existing methods are evaluated.

Retrieval-time interventions. The four published interventions we compare against (§9) span the design space. RECOMP [Xu et al., 2023] compresses retrieved passages. Self-RAG [Asai et al., 2024] delegates retrieve/skip decisions to reflection tokens emitted by a fine-tuned generator. CRAG [Yan et al., 2024] runs a per-passage relevance evaluator and falls back to web search on low-confidence retrievals (we substitute a cross-encoder for the unreleased T5 evaluator and disable the web fallback for matched harness fairness). RAPTOR [Sarthi et al., 2024] builds a hierarchical summary tree over the retrieval corpus offline. Active-retrieval Forward-Looking Active REtrieval generation [Jiang et al., 2023b] interleaves retrieval with generation. None of these papers report cross-dataset τ generalization or multi-scorer faithfulness; our audit is intended to make those checks routine.

Faithfulness measurement. Sentence-level entailment via DeBERTa-v3 NLI [He et al., 2023] is the dominant choice in recent RAG- faithfulness papers. Alternatives include FactScore [Min et al., 2023], AlignScore [Zha et al., 2023], and the RAGAS framework [Es et al., 2023], which uses an LLM judge with an atomic-claim decomposition. We do not benchmark faithfulness scorers in absolute terms; our §6 finding is that *relative agreement* among popular scorers on the same RAG outputs is much weaker than the field's reliance on a single scorer would suggest.

Reproducibility and pre-registration in NLP. Pre-registration is uncommon in NLP relative to clinical or psychological research, but a recent line of work argues for it [van der Lee and Krahmer, 2020, Ulmer et al., 2022]. Rogers et al. [2023] document widespread variance in NLP empirical claims under reproduction, particularly when small samples and unreported tuning are involved. Schaeffer et al. [2023] showed that an apparent emergent-abilities phenomenon dissolved under proper-power evaluation, in a structurally similar episode to the collapse we report in §5. Our audit operationalizes that recommendation for the specific case of RAG faithfulness, with sample sizes, seeds, statistical tests, and effect-size thresholds fixed in machine-checkable logs before any generation runs.

Negative results in machine learning. Workshops and venues for negative results [ACL Workshop on Insights from Negative Results, 2024] have argued that disconfirming empirical claims is a contribution. Our paper sits in that tradition without being a pure negative-results paper: §8 preserves a structural positive finding even after the headline claim collapses.

12 Discussion

What can and cannot be claimed about the refinement paradox. After the audit, the strongest empirical statement supported by the data is: *In short-answer extractive QA on SQuAD with Mistral-7B, aggressive per-passage refinement produces a faithfulness change whose magnitude depends substantially on the choice of faithfulness scorer.* Under DeBERTa-v3 the magnitude is small (0.011), the per-seed contrasts are mostly non-significant, and a matched-similarity intervention finds no causal role for CCS. Under a RAGAS-style LLM-judge the magnitude is large (0.140). The two scorers agree at Pearson $r = 0.18$ on the same outputs. We do not believe a confident claim about the *magnitude* of the paradox is supported by the data; we believe a claim about *measurement instability* is.

Why this matters beyond one phenomenon. The single-scorer / small-sample / in-distribution-tuning pattern that produced the published refinement-paradox magnitude is the modal evaluation pattern across recent RAG-faithfulness papers. If the same audit were applied to the dozen or so recently-reported "faithfulness-improving retrieval interventions," the magnitudes would likely be similarly metric-dependent. We propose pre-registration, multi-scorer reporting, and explicit cross-dataset τ disclosure as the minimum bar for future RAG-faithfulness empirical claims.

The surviving structural finding. §8 preserves a non-trivial result: coherence-preserving uninformative noise produces a smaller faithfulness drop than random off-topic noise at matched per- passage similarity. This says coherence (in some sense, not necessarily CCS) is real and structurally separable from similarity. Future work should look for an operationalization of coherence that survives the matched-similarity test of §4 *and* explains the slope difference of §8. Candidate operationalizations include attention-weighted retrieved-mass [Geva et al., 2023], generator-perplexity over retrieval-set prefixes, or coherence scored by a separate LLM judge with explicit schema.

What HCPC-v2 actually is, after the audit. We retain HCPC-v2 in the released benchmark as a *candidate deployment policy*, not as a controlled coherence-causality probe. Empirically HCPC-v2 is competitive on SQuAD (within 0.0016 of RAPTOR-2L, slightly above CRAG by 0.010) and dominated on HotpotQA (below CRAG by 0.011 on faithfulness, by 2.5 pp on hallucination, and by 18 % on latency). Whether it is worth deploying is a function of the deployment regime; the released LangChain integration makes the policy switchable per query.

Self-RAG's underperformance. The harness fairness question raised in §9 deserves emphasis. Self-RAG was trained on a different distribution than ours and emits reflection tokens that an entailment scorer treats as additional answer text; the apparent 0.57 faithfulness vs. 0.71 for the other methods may substantially be a harness mismatch rather than a real capability gap. We report Self-RAG numbers because they are the matched-harness numbers a downstream practitioner would actually observe if they swapped Self-RAG into a CRAG-style pipeline, which is a useful piece of information even if it is not the fairest possible Self-RAG comparison.

What the audit costs. The full audit ran on free hardware: an Apple M4 Air for Ollama- backed runs and a Kaggle T4×2 session for the 8-bit Self-RAG comparison. Total runtime is well under 25 hours of free Kaggle GPU plus 10–15 hours of M4 wall-clock, and zero paid API tokens. The audit pattern is therefore reproducible by any researcher with free Kaggle access and a workstation, which we view as a prerequisite for the recommendation to make multi-scorer pre-registration routine.

13 Limitations

Single-generator focus on the headline cell. The n=500×5 scaling study (§5) is on SQuAD/Mistral-7B only. Multi-dataset numbers exist at n=30 and n=90 for Mistral, Llama-3, and Qwen-2.5 from earlier rounds and are released alongside this paper, but only the SQuAD/Mistral cell has been re-run at proper power. A reviewer who wanted the same audit on PubMedQA/Llama-3 (a more domain-mismatched setting) would have to re-run that cell themselves. The released Kaggle notebooks make this a 4–6 hour task on a free T4×2.

Long-form QA out of scope. The audit is scoped to short-answer extractive QA. Long-form generation evaluation (MS-MARCO, QASPER) is reported in earlier work as not exhibiting the refinement paradox; our scaling study does not extend to long-form. Long-form RAG faithfulness is an open and harder evaluation problem [Krishna et al., 2021]; we do not address it here.

Free 70B reproduction is budget-blocked. Independent reproduction of the frontier-scale (70B parameter) faithfulness numbers via a non-Groq backend such as Together.ai would require paid API access. Under the zero-dollar constraint of this work that reproduction is blocked. The 70B/120B numbers reported in earlier work via Groq's free tier (Llama-3.3-70B SQuAD paradox ≈ 0.100; GPT-OSS-120B SQuAD paradox ≈ 0.030 on n=30) stand as single-backend observations and should be read with the understanding that we have not yet reproduced them on a second 70B backend. This is disclosed, not papered over.

RAGAS-judge self-bias risk. The RAGAS-style LLM-judge in our §6 multi-scorer audit uses a separate Mistral-7B instance with a different prompt and a structured JSON-schema response. This mitigates direct identity self-bias but not family-level bias: a Mistral-7B judging a Mistral-7B answer may agree with the generated answer's surface plausibility for reasons that have nothing to do with faithfulness. A future version of this audit using a frontier-scale third-party judge (Claude-3.5-Sonnet, GPT-4o, Gemini-2-Pro) would resolve this. Doing so honestly costs paid API tokens, which the zero-dollar constraint of this work disallows.

Self-RAG harness fairness. §9's Self-RAG numbers reflect Self-RAG run inside a CRAG-style harness with a DeBERTa-v3 entailment scorer that does not understand reflection tokens. The "Self-RAG underperforms by ~ 0.14 faithfulness" finding is therefore harness-conditional. A more faithful Self-RAG comparison would require modifying the entailment scorer to strip Self-RAG's control tokens before scoring, or running Self-RAG on the faithfulness benchmark its authors used. We report the matched-harness number because it is the number a downstream practitioner would observe; an aligned-harness number would require additional engineering and is not in this paper's scope.

Causal intervention is single-domain. The Fix 1 matched-similarity intervention runs on SQuAD only, n=200 matched pairs. The same intervention on PubMedQA or HotpotQA is a natural extension that would either replicate or contradict our null. We do not yet know which.

Single retrieval embedder for the scaling study. The scaling study (§5) uses `all-MiniLM-L6-v2` as the embedder. Earlier multi-retriever work showed the paradox is alignment-mismatch-property: the magnitude on SQuAD with stronger embedders (BGE-large, GTE-large) is much smaller than with MiniLM. The collapse at scale that we report is therefore consistent with what we would have predicted from the multi-retriever results, but a full multi- retriever multi-seed multi-metric study at n=500 remains future work.

14 Conclusion

We pre-registered an audit of one prominent RAG faithfulness claim — the refinement paradox — and used it to expose how evaluation choices shape the conclusions that get reported. The published n=30 paradox magnitude collapsed at n=500 × 5 seeds; the proposed causal mechanism (CCS) was not supported by a matched-similarity intervention; the choice of faithfulness scorer moved the magnitude by an order of

magnitude on the same paired outputs; the published threshold τ did not transfer cleanly across datasets; head-to-head comparison against published baselines showed no method dominates.

A structural finding survived: at matched per-passage similarity, coherence-preserving uninformative noise produces a smaller faithfulness drop than random off-topic noise. Coherence is real in some operationalization, even if the published one (CCS) is not the right one.

We release the benchmark, scripts, pre-registrations, and per-query CSVs at github.com/Saket-Maganti/rag-hallucination-detection and DOI 10.5281/zenodo.19757291, on free hardware, so the same audit can be run on any candidate RAG faithfulness intervention before its claims are finalized. We argue that pre-registration, multi-scorer reporting, and explicit cross-dataset τ disclosure should become the minimum bar for empirical claims in RAG faithfulness research.

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